# AI/ML - Driven Maintenance Service Provider Matching Platform for Sri Lanka

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Project Proposal Report

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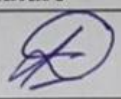
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## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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## ABSTRACT

Online service platforms are increasingly used to connect customers with service providers such as electricians, plumbers, and technicians. However, many existing platforms rely mainly on user ratings and simple listings when recommending service providers. These approaches often fail to verify the actual skills, qualifications, and reliability of professionals, which may lead to poor service quality and reduced user trust.

A major research gap exists in developing systems that evaluate service provider credibility using multiple verification factors instead of relying only on user feedback. In many local platforms, professional credentials, work experience, and historical performance are not properly integrated into the recommendation process.

To address this issue, this research proposes a service provider recommendation system that focuses on dynamic skill verification and credibility evaluation. The proposed component analyzes factors such as certifications, experience, background verification, and service performance to calculate credibility score for each provider. The system will be implemented using technologies including Python, Pandas, NumPy, and Jupyter Notebook for data processing and credibility analysis.

The expected outcome of this research is an improved mechanism for recommending reliable and qualified service providers. The proposed solution can enhance user trust, improve service quality, and support the development of intelligent digital service platforms.

**Keywords:** Credibility Evaluation, Digital Service Platforms, Service Recommendation Systems, Skill Verification, Trust-Based Filtering.

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# TABLE OF CONTENT

DECLARATION ..... 2
ABSTRACT ..... 3
ACKNOWLEDGEMENT ..... 4
TABLE OF CONTENT ..... 5
LIST OF TABLES ..... 7
LIST OF FIGURES ..... 8
LIST OF ABBREVIATIONS ..... 9
LIST OF APPENDICES ..... **Error! Bookmark not defined.**
1 INTRODUCTION ..... 10
1.1 Background and Context ..... 10
1.2 Literature Review ..... 11
1.3 Research Gap ..... 12
2 OBJECTIVE ..... 14
2.1 Main Objective ..... 14
2.2 Specific Objectives ..... 14
3 METHODOLOGY ..... 16
3.1 Research Approach ..... 16
3.2 Technical Methods and Algorithms ..... 16
3.3 Tools and Platforms with Justification ..... 17
3.4 Data Collection and Ethical Considerations ..... 18
3.5 Validation Strategy and Performance Metrics\ ..... 19
4 HIGH-LEVEL SYSTEM ARCHITECTURE ..... 20
4.1 Overview of Proposed Solution and its Components ..... 20
4.2 System Diagram of Components ..... 20
4.2.1 System Architecture Overview ..... 22
4.2.2 Layered Functional Analysis ..... 22
4.3 Highlight of Individual Component and its Parts ..... 23

4.4 Component Interactions ..... 24
4.5 Data Flow Explanation ..... 24
4.6 Scalability and Security Considerations ..... 26
5 USER REQUIREMENTS ..... 27
5.1 Stakeholder Identification ..... 27
5.2 Functional Requirements ..... 27
5.3 Non-Functional Requirements ..... 28
6 COMMERCIALIZATION PLAN ..... 30
6.1 Target Market and Customer Persona ..... 30
6.2 Value Proposition ..... 30
6.3 Revenue Model ..... 30
6.4 Cost Estimation ..... 31
6.5 Pricing Strategy ..... 31
6.6 Competitive Advantage ..... 31
6.7 Intellectual Property (IP) Considerations ..... 31
7 BUDGET AND JUSTIFICATION ..... 32
7.1 Estimated Budget Breakdown ..... 32
7.2 Detailed Justification for Each Expense ..... 32
7.2.1 Hardware Costs ..... 32
7.2.2 Software Licenses ..... 32
7.2.3 Cloud Services ..... 32
7.2.4 Connectivity and Research Access ..... 33
7.2.5 Web Hosting and Domain ..... 33
8 WORK BREAKDOWN STRUCTURE (WBS) ..... 34
9 GANTT CHART ..... 36
10 REFERENCES LIST ..... 37
11 APPENDICES ..... **Error! Bookmark not defined.**

## LIST OF TABLES

Table 1: List of Abbreviations..... 9
Table 2: Summary of Tools and Platforms..... 17
Table 3: Summary of Estimated Budget ..... 32

## LIST OF FIGURES

Figure 1: System Architecture Diagram ..... 21
Figure 2: Work Breakdown Structure ..... 34
Figure 3: Member-wise Work Breakdown Structure ..... 35
Figure 4: Gantt Chart ..... 36

# LIST OF ABBREVIATIONS

*Table 1: List of Abbreviations*

| Abbreviation | Full Form |
|---|---|
| AI | Artificial Intelligence |
| API | Application Programming Interface |
| DB | Database |
| DSR | Design Science Research |
| FR | Functional Requirement |
| ML | Machine Learning |
| NFR | Non-Functional Requirement |
| NLP | Natural Language Processing |
| SQL | Structured Query Language |
| UI | User Interface |
| UX | User Experience |

INTRODUCTION

## 1.1 Background and Context

With the rapid growth of digital technologies, online service platforms have become an important medium for connecting customers with service providers. In many countries, digital marketplaces allow users to easily find professionals such as electricians, plumbers, technicians, and other maintenance service providers through mobile applications or web platforms. These platforms help reduce the time and effort required to locate reliable service providers while improving service accessibility for customers.

However, in the Sri Lankan context, many households and small businesses still face significant challenges when trying to find trustworthy and skilled service providers. Most customers rely on informal methods such as personal recommendations, social media posts, or classified advertisement websites to locate technicians. While these methods may sometimes be convenient, they do not guarantee the reliability, professional qualifications, or credibility of the service provider. As a result, customers may experience poor service quality, financial losses, or safety risks due to unqualified technicians.

One of the major issues associated with existing service platforms is the lack of a structured verification mechanism to evaluate the skills and credibility of service providers. In many current systems, providers can easily register and advertise their services without proper verification of their professional certifications, work experience, or background information. Consequently, customers often find it difficult to determine whether a service provider is truly qualified to perform the required task.

Existing digital platforms mainly rely on user ratings and reviews to evaluate service providers. Although ratings can provide useful feedback about customer experience, they may not accurately represent the actual competence or reliability of the provider. Reviews can sometimes be biased, manipulated, or limited in number, which reduces the effectiveness of rating-based evaluation systems. Therefore, relying solely on ratings may not be sufficient to ensure trustworthy service recommendations.

Recent advancements in intelligent systems and data-driven technologies such as machine learning and recommender systems have created new opportunities to improve service recommendation platforms. By analyzing multiple credibility indicators such as professional certifications, years of experience, service completion history, and customer feedback, intelligent systems can generate more reliable recommendations for users. These approaches help reduce uncertainty and increase user trust when selecting service providers.

Therefore, there is a growing need to develop an intelligent system that can verify the skills and credibility of service providers before recommending them to users. Such a system can significantly improve the reliability and transparency of online service marketplaces while ensuring better service quality for customers.

This research proposes the development of a multi-component intelligent service provider recommendation system. Within this system, Component 3 focuses on dynamic skill verification and credibility evaluation of service providers**.** This component analyzes multiple factors including provider certifications, work experience, background verification status, and historical service performance to generate a credibility score. Based on this score, the system filters and ranks service providers to ensure that only qualified and trustworthy professionals are recommended to users.

## 1.2 Literature Review

Online service platforms have become increasingly popular as digital solutions for connecting customers with service providers. These platforms help users search for professionals who can perform various services such as home maintenance, repair work, and technical support. Most modern systems utilize recommended systems to match user requirements with suitable service providers.

Traditional recommendation systems commonly rely on user ratings and reviews to evaluate the quality-of-service providers. Ratings provide useful feedback from previous customers and can help users make informed decisions when selecting professionals. However, many studies highlight that rating-based evaluation alone is insufficient for measuring the true competence and reliability of service providers.

Research in intelligent recommendation systems suggests that trust-based evaluation models can significantly improve recommendation accuracy. These models consider multiple credibility factors such as professional certifications, years of experience, job completion history, and service reliability. By analyzing such factors, the system can better assess whether a service provider is capable of delivering high-quality services.

Another important concept discussed in recent research is credibility scoring. Credibility scoring methods combine different indicators of trust and performance to generate a reliability score for each provider. This approach helps platforms filter out unqualified providers and prioritize those with proven professional competence.

Furthermore, studies in machine learning-based recommendation systems demonstrate that analyzing historical service data can improve the prediction of provider performance. Factors such as customer satisfaction levels, service completion rates, and response times can be used to evaluate the consistency and reliability of service providers.

### 1.3 Research Gap

Although many digital service platforms and recommendation systems have been developed, several limitations still exist in the way service providers are evaluated and recommended. Most existing systems mainly rely on customer ratings and reviews to measure service quality. However, ratings alone may not accurately represent the professional competence or trustworthiness of the service providers.

Another limitation is the lack of mechanisms to verify professional credentials such as certifications, work experience, or background information. Without proper verification, unqualified or unreliable providers may appear in recommendation results, which can negatively affect user trust and satisfaction.

Furthermore, many current systems do not integrate multiple credibility indicators into a single evaluation framework. Important factors such as professional qualifications, service completion history, and provider reliability are often ignored or only partially considered during the recommendation process.

In the Sri Lankan context, research on intelligent service provider verification systems is still limited. Most local platforms operate as basic service directories rather than intelligent systems that analyze provider credibility and performance.

Therefore, there is a clear need for a system that can evaluate service providers using multiple credibility indicators and generate reliable recommendations. This research aims to address this gap by introducing dynamic skill verification and credibility evaluation mechanism within a multi-stage service recommendation system.

# 2 OBJECTIVE

The objectives of this research are designed to transform a manual, fragmented maintenance service market into an automated, data-driven ecosystem. Each objective is formulated to be SMART (Specific, Measurable, Achievable, Relevant, and Time-bound).

## 2.1 Main Objective

The primary goal of this research is to design, develop, and evaluate a robust, AI-driven web-based platform tailored for the Sri Lankan maintenance service sector. The system aims to transform the current manual and inefficient service discovery process into an automated, data-driven ecosystem by implementing a sequential four-stage intelligent pipeline. By fusing deep semantic intent recognition with real-time contextual factors and credibility verification, the platform seeks to achieve a high recommendation precision of 88%, thereby enhancing user trust and operational efficiency in the local marketplace.

**Key Pillars of the Main Objective:**

- **Automation of the Service Lifecycle:** To eliminate the manual effort required in traditional directories by automating the path from a natural language user request to the final expert recommendation.
- **Intelligent Skill Identification:** To leverage advanced NLP architectures like BERT to bridge the semantic gap in mixed-language (Sinhala-English) service requests, ensuring highly accurate provider matching.
- **Verification and Trust Integration:** To build a transparent marketplace by integrating dynamic credibility scoring and aspect-level sentiment analysis, ensuring that recommendations are based on verified skills and customer satisfaction.
- **Optimized Web Accessibility:** To deliver a high-performance web application accessible via a RESTful API architecture, providing a seamless user experience even in low-bandwidth network conditions common in Sri Lanka.

## 2.2 Specific Objectives

- To verify the professional qualifications and certifications of service providers.

- To evaluate the credibility of service providers using background verification and experience data.
- To analyze historical service performance based on customer ratings and job completion records.
- To generate a credibility score that enhances the accuracy and trustworthiness of service provider recommendations.

# 3 METHODOLOGY

## 3.1 Research Approach

This study follows a Design Science Research (DSR) approach, which focuses on designing and developing technological artifacts to solve real-world problems. The main goal of this research is to develop a component that verifies the skills and credibility of service providers within an intelligent service recommendation system.

The Design Science approach is suitable for this research because it allows the development of a practical solution in the form of a system component that improves the reliability of service provider's recommendations. In this research, Component 3 is designed as a dynamic skill verification and credibility evaluation mechanism that analyzes several factors such as professional certifications, experience, background verification, and historical service performance.

The research process includes several stages such as problem identification, system design, implementation of the credibility evaluation model, and validation of the system's performance. Through this approach, the proposed component contributes both a practical solution for improving service provider verification and an academic contribution to intelligent recommendation systems.

## 3.2 Technical Methods and Algorithms

Component 3 focuses on verifying the credibility and professional competence of service providers before they are recommended to users. To achieve this, the system uses a multi-factor credibility scoring model that evaluates service providers based on several indicators.

The technical method involves calculating a credibility score for each provider using weighted factors. These factors include professional certifications, years of experience, background verification status, customer ratings, and job completion history.

A scoring algorithm is used to combine these factors into a single credibility value. The credibility score is calculated using a weighted formula:

Credibility Score = (0.4 × Credential Score) + (0.3 × Performance Score) + (0.3 × Trust Score)

Where:

Credential Score represents the verification of professional certifications and qualifications.

Performance Score represents historical performance indicators such as customer ratings and job completion rates.

Trust Score represents reliability indicators such as background verification and service history.

After calculating the credibility score, service providers are ranked based on their scores, and those with higher credibility are prioritized in the recommendation process. This method ensures that the system recommends only trustworthy and qualified professionals.

### 3.3 Tools and Platforms with Justification

*Table 2: Summary of Tools and Platforms*

| **Tool / Platform** | **Purpose** | **Justification** |
|---|---|---|
| Python | To implement the credibility scoring model and perform data analysis. | Python is widely used in data science and machine learning due to its simplicity and strong ecosystem of libraries for data processing and algorithm development. |
| Pandas | To process and manage structured datasets of service provider information and performance data. | Pandas provides efficient data manipulation and analysis capabilities, making it suitable for handling large datasets. |

| | | |
|---|---|---|
| NumPy | To perform numerical computations required for credibility score calculations. | NumPy supports fast mathematical operations and array processing, which improves computational efficiency. |
| Jupyter Notebook | To develop, test, and visualize the algorithm and experimental results. | Jupyter Notebook provides an interactive development environment that allows easy testing, debugging, and visualization of data analysis processes. |
| MySQL / MongoDB | To store service provider information, verification records, and historical performance data. | These database systems provide reliable data storage and efficient retrieval, which are essential for managing large datasets in recommendation systems. |

### 3.4 Data Collection and Ethical Considerations

The dataset used in this research will consist of service provider information and historical service records. The data may include attributes such as provider qualifications, years of experience, customer ratings, job completion rates, and verification status.

Data may be collected from publicly available datasets, simulated datasets created for experimental purposes, or anonymized platform data. During data collection, personal information of service providers and customers will be protected to ensure privacy and confidentiality.

Ethical considerations are an important aspect of this research. Any personal identifiers such as names, phone numbers, or addresses will be removed or anonymized before analysis. The study will follow responsible data usage practices and ensure that the collected data is used only for research purposes.

Additionally, the system will be designed to ensure fairness and avoid biased evaluation of service providers. The credibility scoring mechanism will be based on objective performance indicators rather than personal or demographic attributes.

### 3.5 Validation Strategy and Performance Metrics\

To evaluate the effectiveness of the proposed Component 3, a validation strategy will be implemented using experimental testing and performance evaluation metrics.

The system will be tested by applying the credibility scoring model to a dataset of service providers. The performance of the model will be evaluated by analyzing whether the system successfully prioritizes trustworthy and qualified providers.

Several performance metrics will be used to assess the effectiveness of the credibility evaluation model. These include:

- **Accuracy** – Measures how accurately the system identifies credible service providers.
- **Precision** – Measures the proportion of recommended providers who are truly reliable and qualified.
- **Recall** – Measures how effectively the system identifies all relevant trustworthy providers within the dataset.
- **F1 Score** – Combines precision and recall providing a balanced evaluation of the system's performance.

Additionally, ranking evaluation metrics may be used to measure how well the system prioritizes high-quality service providers within the recommendation list.

Through these validation techniques, the research will assess whether the proposed credibility evaluation component improves the reliability and quality of service provider recommendations.

# 4 HIGH-LEVEL SYSTEM ARCHITECTURE

## 4.1 Overview of Proposed Solution and its Components

The proposed solution is a centralized, AI-driven web platform designed to automate the maintenance service provider matching process in Sri Lanka. The system operates through a four-stage intelligent pipeline:

- **Component 1 (Matching Engine):** Fuses BERT-based semantic intent with behavioural data.
- **Component 2 (Contextual Module):** Refines results based on proximity and real-time environment.
- **Component 3 (Skill Verification):** Quantifies trust through credential cross-referencing.
- **Component 4 (Sentiment Analysis):** Finalizes rankings using aspect-level review evaluation.

## 4.2 System Diagram of Components

The following diagram illustrates the interaction between the frontend interfaces, the modular ML pipeline, and the dual-database storage layer.

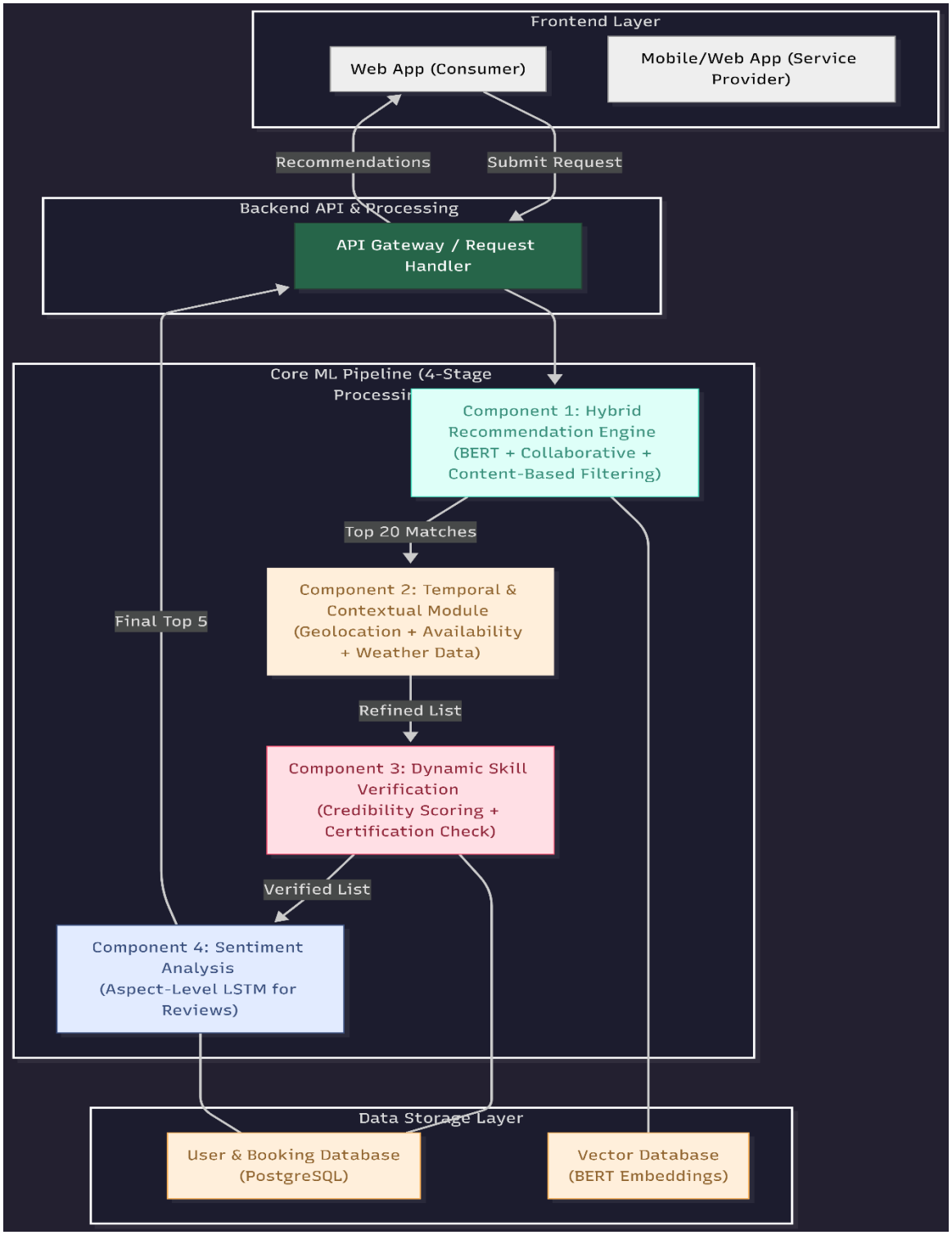



*Figure 1: System Architecture Diagram*

### 4.2.1 System Architecture Overview

The proposed architecture is designed as a modular, high-performance web-based ecosystem that bridges the gap between consumers and service providers through a sequential **4-Stage Core ML Pipeline**. The system is logically divided into three primary layers: Frontend, Backend, and Data Storage.

### 4.2.2 Layered Functional Analysis

- **Frontend Layer:** The system utilizes a dual-interface approach with a Web App (Consumer) for submitting service requests and a Web App (Service Provider) for managing jobs.
- **Backend API & Processing:** The API Gateway / Request Handler acts as the central controller, receiving natural language requests from the consumer and routing them through the ML pipeline before returning the final "Top 5" recommendations.
- **Core ML Pipeline (The 4-Stage Process):**
    1. **Component 1 (Hybrid Recommendation):** The initial filter uses BERT for semantic understanding along with Collaborative and Content-Based **Filtering** to generate the "Top 20 Matches" by interacting with a high-dimensional Vector Database.
    2. **Component 2 (Temporal & Contextual Module):** This module refines the list by evaluating real-time Geolocation, Availability, and Weather Data, ensuring the recommendations are contextually relevant.
    3. **Component 3 (Dynamic Skill Verification):** It performs a rigorous credibility check using Certification Verification and historical Credibility Scoring to produce a "Verified List."
    4. **Component 4 (Sentiment Analysis):** The final stage utilizes an Aspect-Level LSTM model to analyse past reviews, selecting the most reliable candidates to present as the "Final Top 5."

- **Data Storage Layer:** The architecture integrates a PostgreSQL relational database for structured user and booking history, alongside a specialized Vector Database for storing and searching BERT-generated embeddings.

### 4.3 Highlight of Individual Component and its Parts

Component 3- Dynamic skill verification & credibility focuses on verifying the professional competence and credibility of service providers before they are recommended to users. This component evaluates several factors related to provider qualifications and past performance.

Component 3 consists of the following submodules:

- **Credential Verification Module**
  This module verifies whether the service provider possesses valid professional certifications and relevant qualifications required to perform the service.

- **Experience Evaluation Module**
  This module evaluates the provider's years of experience and service history to assess professional competence.
- **Trust and Background Verification Module**
  This module checks reliability indicators such as background verification status and provider authenticity.
- **Performance Analysis Module**
  This module analyzes historical performance data such as customer ratings, job completion rates, and service reliability.
- **Credibility Scoring Module**
  This module combines the outputs of the previous modules and calculates a final credibility score used to rank service providers.

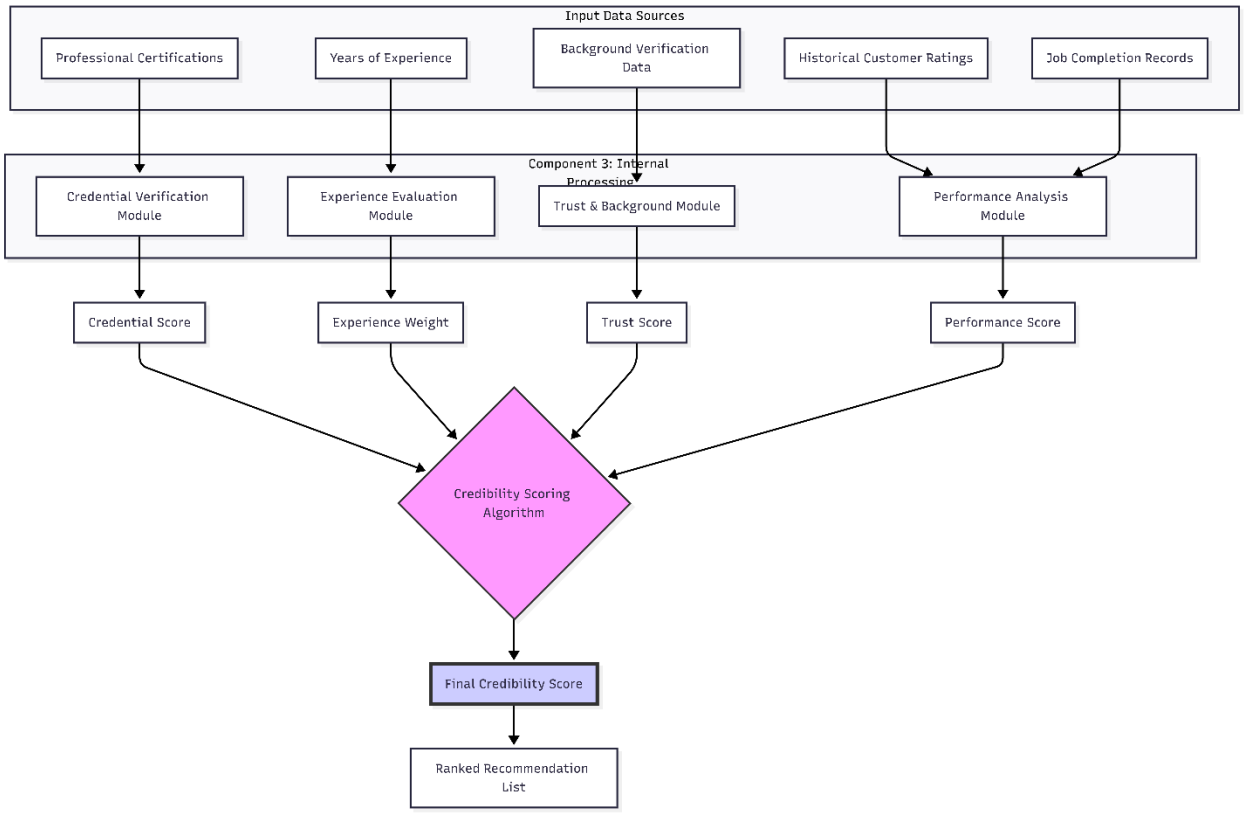



*Figure 2: Internal Architecture & Data Flow*

### 4.4 Component Interactions

The system follows a pipe-and-filter architecture:

1. The API Gateway receives a request and triggers Component 1.
2. Component 1 queries the Vector Database and passes the "Top 20" to Component 2.
3. Component 2 applies spatio-temporal filters and passes the refined list to Component 3.
4. Component 3 validates credentials against the Relational Database before Component 4 performs the final sentiment ranking.

### 4.5 Data Flow Explanation

The data flow within the system begins when the user submits a service request through the platform interface. The system retrieves service provider data from the database, including provider qualifications, service categories, historical performance records, and user reviews.

This data is processed sequentially by each system component. Component 1 filters providers based on service category. Component 2 applies contextual filters such as location and availability. Component 3 then evaluates provider credibility using verification and performance analysis. The processed data is then passed to Component 4 for review analysis.

Finally, the system generates a ranked list of service providers based on relevance, credibility, and customer feedback, which is presented to the user as the final recommendation.

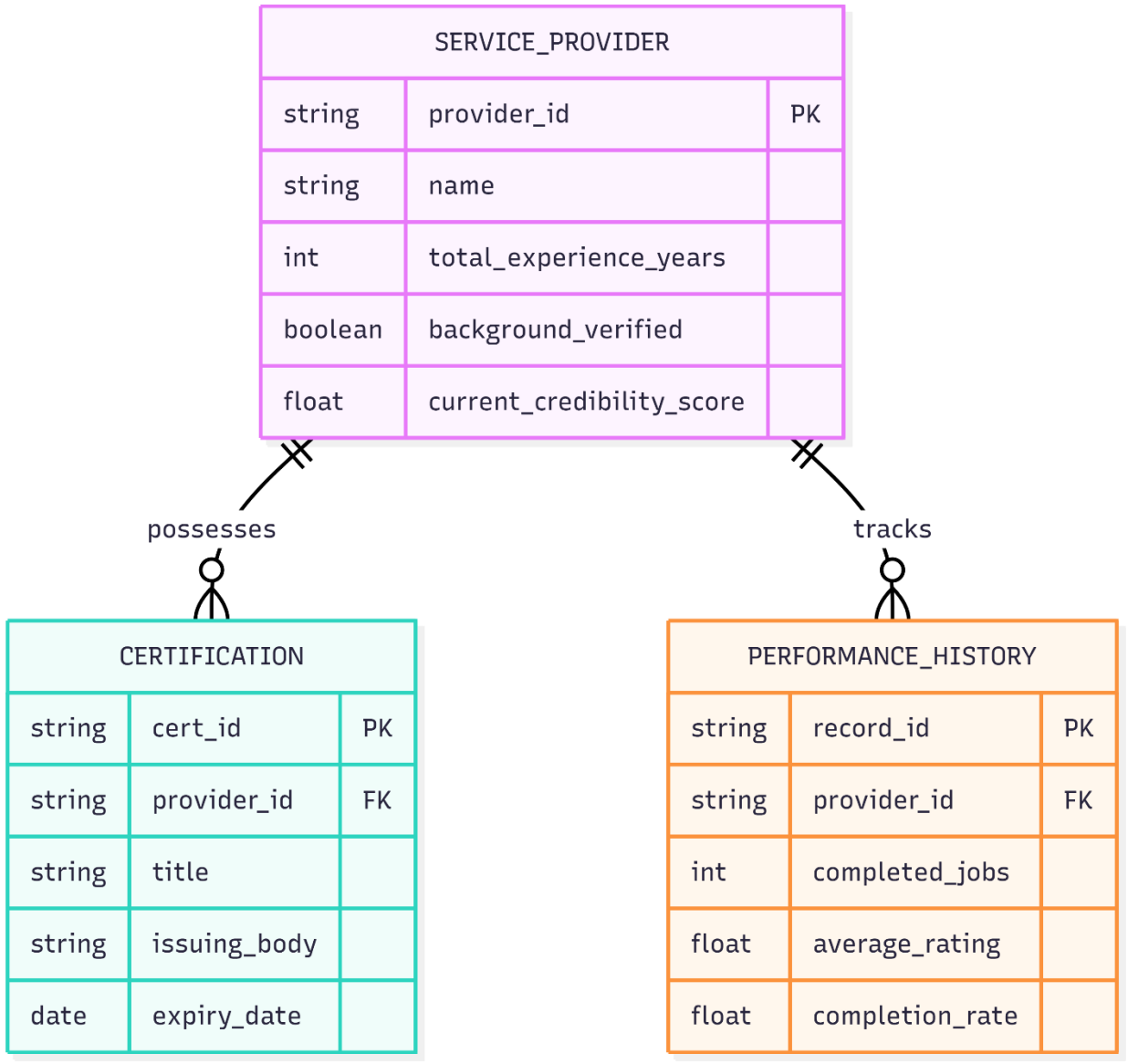



*Figure 3: Database Schema for Component 3*

### 4.6 Scalability and Security Considerations

The proposed system is designed to support scalability and secure data management. As the number of users and service providers increases, the system architecture can be scaled by using cloud-based infrastructure and distributed database systems to handle larger volumes of data efficiently.

Security measures are also important to ensure the protection of user and service provider information. Sensitive data such as personal details and verification records will be securely stored in the database with appropriate access controls. Authentication mechanisms will be used to ensure that only authorized users and administrators can access or modify system data.

Additionally, data anonymization techniques will be applied when analyzing service provider performance data to protect individual privacy and maintain ethical research standards.

# 5 USER REQUIREMENTS

## 5.1 Stakeholder Identification

The proposed system involves several stakeholders who interact with or benefit from the service provider recommendation platform.

- **Customers / Service Seekers**
  Customers are the primary users of the system. They use the platform to search for reliable service providers such as electricians, plumbers, or technicians based on their service needs. Customers expect accurate recommendations and trustworthy professionals.
- **Service Providers**
  Service providers are professionals who register on the platform to offer their services. They provide their professional information, certifications, and service history so that the system can evaluate their credibility and match them with suitable customers.
- **System Administrators**
  Administrators manage and maintain the platform. Their responsibilities include verifying service provider information, monitoring system performance, managing user accounts, and ensuring the reliability of the platform.
- **Researchers / System Developers**
  Researchers and developers design and improve the system components, including the credibility evaluation mechanism. They analyze system performance and update algorithms to improve recommendation accuracy.

## 5.2 Functional Requirements

The functional requirements describe the main functions that the system must perform.

FR1: The system shall allow users to submit service requests based on their required service type.

FR2: The system shall allow service providers to register and submit their professional information, including certifications and experience.

FR3: The system shall verify the credentials and qualifications of registered service providers.

FR4: The system analyzes historical service performance data such as ratings and job completion records.

FR5: The system shall calculate credibility scores for each service provider using multiple evaluation factors.

FR6: The system shall rank service providers based on their credibility scores.

FR7: The system shall recommend the most reliable service providers to users.

FR8: The system shall store and manage service provider information in the system database.

FR9: The system shall allow administrators to monitor and manage system data and verification processes.

### 5.3 Non-Functional Requirements

The non-functional requirements describe the quality attributes that the system must satisfy.

- **Security**
  The system must ensure that sensitive user and service provider information is securely stored and protected from unauthorized access. Authentication and access control mechanisms should be implemented.
- **Performance**
  The system should provide service provider recommendations within a short response time to ensure a smooth user experience.
- **Usability**
  The platform should have a simple and user-friendly interface so that users can easily search for service providers and view recommendations.
- **Reliability**
  The system should provide accurate and consistent recommendations based on verified data and performance analysis.

- **Scalability**

  The system should be able to handle an increasing number of users and service providers as the platform grows.
- **Data Privacy**

  Personal information of users and service providers must be handled according to ethical data usage practices, ensuring confidentiality and privacy.

# 6 COMMERCIALIZATION PLAN

## 6.1 Target Market and Customer Persona

Understanding the "Human" side of the system is vital for the 88% precision target.

- **Target Market:** The focus is on the Urban Maintenance Sector in Sri Lanka. Currently, this market is fragmented, relying on word-of-mouth or unverified social media posts.
- **Customer Persona - The Tech-Savvy Urbanite:** This user lives in Colombo or Gampaha, uses a smartphone for everything, but fears being "ripped off" by unverified workers. They need a **Web Platform** that acts as a middleman for trust.
- **Provider Persona - The Skilled Freelancer:** Skilled individuals who lack marketing budgets. Your platform gives them a "Credibility Score" based on their actual skills, levelling the playing field against large maintenance companies.

## 6.2 Value Proposition

The novelty lies in the Multi-Stage Refinement.

- **Precision as a Product:** While competitors offer search, you offer matching. The value is in the "Top 5" list which saves the user from calling 10 different people to find one who is available.
- **Bilingual Intelligence:** BERT allows users to type in "Singlish" or Sinhala, which is the primary mode of communication in local maintenance, making the platform more inclusive than English-only apps.

## 6.3 Revenue Model

- **Transaction Fee:** A success-based fee. This ensures the platform's interests are aligned with the user (money is only made if a good match happens).
- **Data Insights (B2B):** Aggregated, anonymized data on service demand patterns (e.g., "Which areas need more plumbers?") can be sold to training institutes or tool manufacturers.

### 6.4 Cost Estimation

- **Infrastructure:** High-performance GPU instances (like AWS p3) for the initial training of the LSTM and BERT models.
- **Data Acquisition:** Costs associated with scraping or purchasing local service datasets to solve the "Cold Start" problem in Recommendation Engines.
- **Hosting:** Monthly costs for PostgreSQL and Vector DB (Pinecone) to maintain the 2.0s latency requirement.

### 6.5 Pricing Strategy

- **Freemium for Providers:** Basic listing is free; a small monthly fee for a "Verified Badge" (which requires Component 3 verification).
- **Dynamic Pricing:** In the future, the platform could suggest higher prices during high-demand periods (e.g., AC repair during heatwaves), benefiting the providers while taking a commission.

### 6.6 Competitive Advantage

- **Semantic Advantage:** Unlike ikman.lk, your system understands that "pipe ekak hayi karanna" means "plumbing."
- **Credibility Advantage:** Most platforms are "pay-to-play" (whoever pays more is at the top). Your system is Performance-Based, ranking people by sentiment and verified history.

### 6.7 Intellectual Property (IP) Considerations

- **Software Copyright:** The source code of the **FastAPI** backend and the specific **BERT-LSTM** fusion logic.
- **Database Rights:** The unique dataset of Sinhala-English service intents and the "Vector Embeddings" of local providers.
- **Patenting:** The specific 4-stage pipeline logic could be considered for a "Business Process Patent" if it shows a significant leap over existing state-of-the-art systems.

# 7 BUDGET AND JUSTIFICATION

## 7.1 Estimated Budget Breakdown

*Table 3: Summary of Estimated Budget*

| Category | Item Description | Estimated Cost (LKR) |
|---|---|---|
| Hardware | Existing Personal Computer (Laptop/Desktop) | 0 (Existing) |
| Software | Open Source Libraries (FastAPI, PyTorch, BERT) | 0 (Free) |
| Cloud Services | Google Colab (Free Tier) & Render/Railway Free Tier | 0 (Free) |
| Connectivity | High-speed Internet for Model Training & Research | 15,000 |
| Software/Web | Domain Name (.com/.lk) & Basic Hosting (Annual) | 8,000 |
| Documentation | Printing, Binding, and Thesis Preparation | 7,000 |
| Total | | 30,000 |

## 7.2 Detailed Justification for Each Expense

### 7.2.1 Hardware Costs

- **Justification:** As this is an academic research project, the primary hardware used will be our existing personal computer.

### 7.2.2 Software Licenses

- **Justification:** All software frameworks, including Python, FastAPI, and BERT, are Open Source, meaning there are no licensing costs involved.

### 7.2.3 Cloud Services

- **Justification:** To train the BERT and LSTM models, Google Colab **(**Free Tier**)** will be utilized, providing access to high-performance GPUs at no cost. For the final web application hosting, platforms with a free entry tier (e.g., Vercel) will be used to demonstrate the prototype.

#### 7.2.4 Connectivity and Research Access

- **Justification:** Consistent high-speed internet is essential for downloading large pre-trained models (BERT-base) and accessing IEEE/ACM research databases. This budget covers the specialized data requirements for the project duration.

#### 7.2.5 Web Hosting and Domain

- **Justification:** To evaluate the User Requirements, a live web URL is needed. A nominal fee is allocated for professional domain and basic web hosting to make the system accessible to stakeholders for testing.

# 8 WORK BREAKDOWN STRUCTURE (WBS)

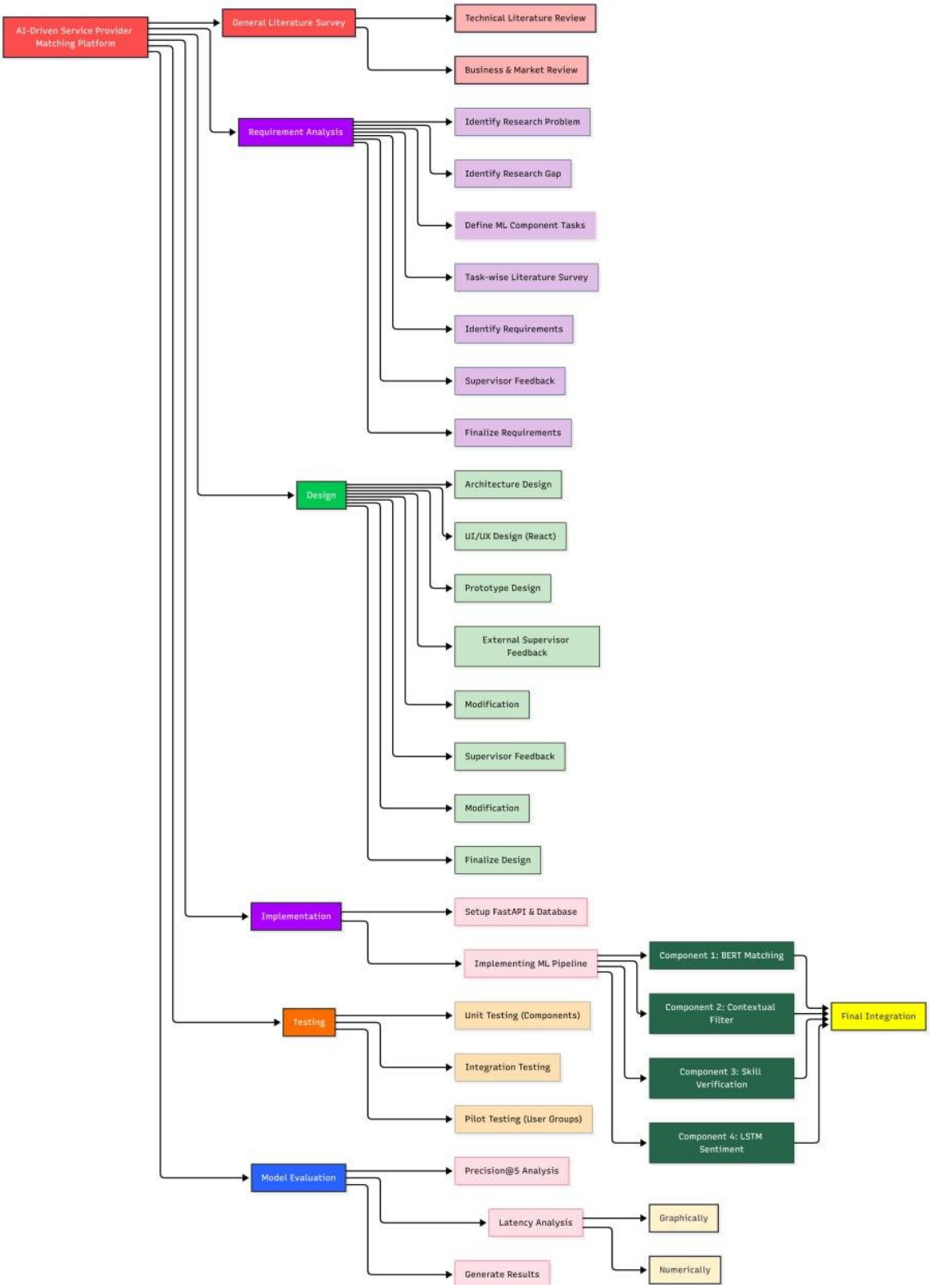



*Figure 2: Work Breakdown Structure*

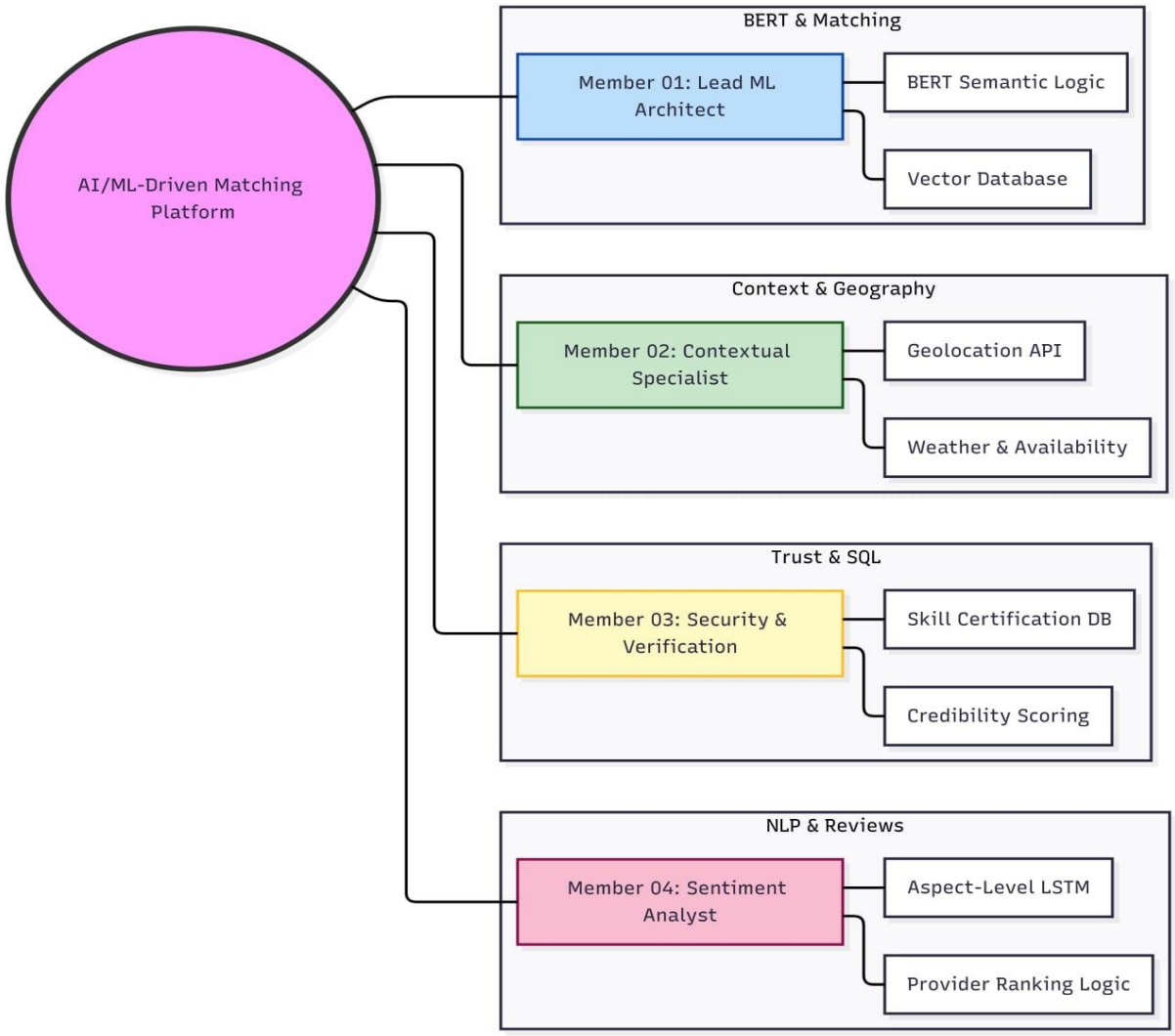



*Figure 3: Member-wise Work Breakdown Structure*

# 9 GANTT CHART

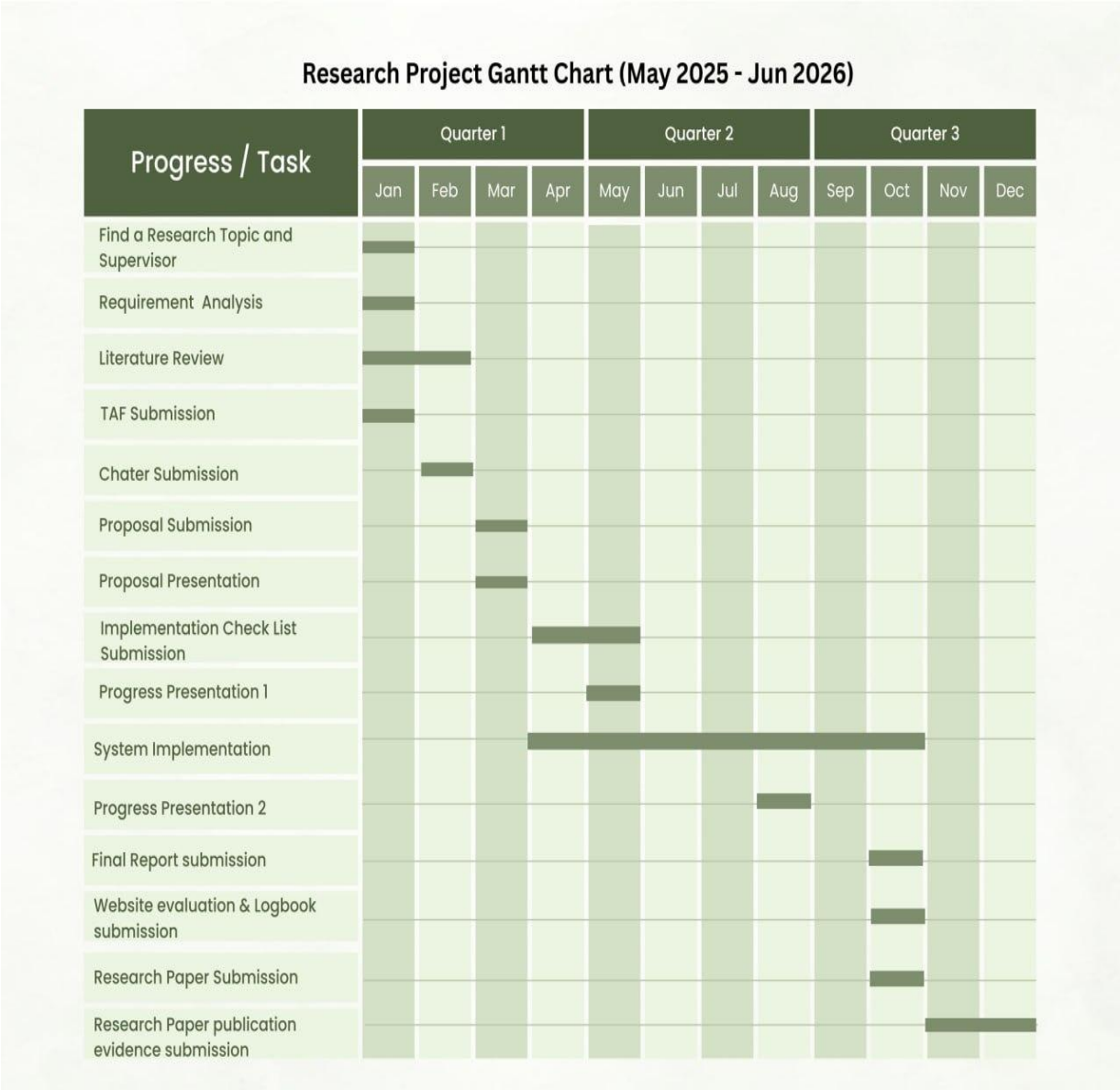



*Figure 4: Gantt Chart*

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